1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlighted that inflammatory/immune readouts (HMGB1, CHI3L1, and a nine-protein signature) are prominent in TNBC, yet robust immune organization (e.g., TLS/TILs) does not always translate into effective anti-tumor control. Why do some immune-infiltrated TNBCs fail to achieve durable control? 2) Tier 1 paths used: I built primarily on the HMGB1 (Core Path 2) and CHI3L1 (Core Path 3) axes because both are secreted inflammatory mediators/alarmins tied to EMT, matrix remodeling, and protumor inflammation, which mechanistically can disable local adaptive immunity. I considered the nine-protein proteomic signature (Core Path 1) as contextual but kept it out of the core claim to preserve mechanistic clarity. 3) Tier 2 support: Proliferation and risk-score signals (Ki-67 / RS), cancer-stem-cell markers, EMT/ECM effectors (vimentin, MMP9) and the 5q15 genomic context provide plausible modifiers—e.g., high proliferation/CSC or active ECM remodeling could amplify HMGB1/CHI3L1-driven myeloid recruitment and TLS dysfunction. 4) Why PFI endpoint: Progression-free interval is selected because the central question concerns failure of local immune control to prevent recurrence/progression (PFI is more sensitive than OS to early immune-related control/relapse signals in TNBC within TCGA). 5) Deliberate exclusions: I excluded socioeconomic/external-context nodes (education), distant/systemic biomarkers not mechanistically tied to local TLS function (t-tau, CK), and folded the composite nine-biomarker signature into secondary_modifiers rather than the core claim to avoid conflating descriptive panels with mechanistic drivers?

Working Hypothesis

In triple-negative breast cancer (TNBC), the presence of TLS-like lymphoid aggregates is not invariably associated with durable tumor control if tumor-derived alarmins HMGB1 and CHI3L1 are highly expressed. The expanded hypothesis posits that high HMGB1 and CHI3L1 reprogram the TLS microenvironment through recruitment and polarization of myeloid cells (e.g., MDSCs, M2-like TAMs), upregulation of ECM remodeling (e.g., MMP9, vimentin) and suppression of antigen presentation (diminished dendritic cell maturation and HLA class II activity). This creates a dysfunctional TLS that supports local immune evasion, T cell exhaustion, and early relapse, leading to shorter progression-free interval (PFI) despite TLS presence. In TLS-rich, HMGB1/CHI3L1-low tumors, TLSs remain functionally competent with effective antigen presentation and T cell activation, correlating with longer PFI. Secondary modifiers (Ki-67/proliferation, cancer stem cell markers, EMT/ECM effectors, and 5q15 genomic context) modulate the strength of HMGB1/CHI3L1-driven TLS dysfunction and thus influence the magnitude of PFI differences. Therapeutically, disrupting HMGB1/CHI3L1 signaling or downstream myeloid recruitment should restore TLS functionality and improve anti-tumor immunity, offering a mechanistic and clinically testable avenue to convert TLS presence into durable control.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	HMGB1, CHI3L1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

HMGB1: HMGB1 (Core Path 2) and CHI3L1 (Core Path 3) axes because both are secreted inflammatory mediators/alarmins tied to EMT, matrix remodeling, and protumor inflammation, which mechanistically can disable local adaptive immunity.

CHI3L1: CHI3L1 (Core Path 3) axes because both are secreted inflammatory mediators/alarmins tied to EMT, matrix remodeling, and protumor inflammation, which mechanistically can disable local adaptive immunity.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Biomarker:nine final biomarkers	Supporting modifier from knowledge graph
Secondary	MMP9	Supporting modifier from knowledge graph
Secondary	VIM (vimentin)	Supporting modifier from knowledge graph
Secondary	Ki-67	Supporting modifier from knowledge graph
Secondary	cancer stem cell marker	Supporting modifier from knowledge graph
Secondary	5q15	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analyses of TNBC cohorts with TLS characterization, HMGB1/CHI3L1 expression, and PFI to test the TLS-high HMGB1/CHI3L1-low vs -high split;

Prediction 2: spatial and molecular profiling to assess TLS architecture, DC maturation, T cell activation, and myeloid infiltration;

Prediction 3: in vitro co-culture systems with TNBC cells engineered to modulate HMGB1/CHI3L1 to evaluate effects on DC maturation and T cell priming in TLS-like microenvironments;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Biomarker:nine final biomarkers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MMP9: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.